

Master’s Thesis Proposal

Harvard Master’s in Data Science

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1. Title

Trusted-Weak Concept Transfer: An Empirical Study of Inference-Time Scalable Oversight in Large Language Models.

2. Advisor

Adityanarayanan (Adit) Radhakrishnan — Assistant Professor, MIT Department of Mathematics; Associate Member, Broad Institute of MIT and Harvard (aditradha.com). A Harvard SEAS co-advisor will be added if the program requires an internal sponsor when the primary advisor is external; pending confirmation with Daniel Weinstock.

3. Abstract

Scalable oversight asks how a less-capable supervisor (a smaller AI system or a human aided by one) can meaningfully evaluate the behavior of a more-capable system. Existing approaches are mostly *evaluation-time* — sandwiching, debate, AI-assisted critique — and operate on the model’s outputs. This thesis proposes and empirically tests a complementary *inference-time* oversight protocol: a smaller, more interpretable model W extracts a concept direction from its own internal representations; the direction is transferred to a larger, less interpretable model S ; the transferred direction is then used to steer S at decode time; and W verifies S ’s steered outputs post-hoc. The protocol provides a closed loop in which the weaker, trusted model both *defines* the target behavior and *checks* that the stronger model has exhibited it. I evaluate three transfer protocols (prompt curation, activation correlation, probe-as-supervisor) across three model families spanning $\sim 8\times$ in parameter count, on four concept families skewed toward safety (refusal, sycophancy, truthfulness, sentiment), with a pre-registered decision gate before the more expensive runs. The mechanistic chapter uses my advisor’s Average Gradient Outer Product (AGOP) framework to characterize the concept-relevant input subspace in W and S , and asks whether cross-model AGOP subspace alignment predicts transfer success. A separate summer paper on Mixture-of-Experts routing-bias steering (Attention-Guided Router Steering, AGRS) is the proof-of-concept that motivates this thesis.

4. Motivation and Background

Prior research and current work. My prior project at Harvard was on hyperbolic prototype embeddings for hierarchical artistic style,¹ which trained matched-dimension Euclidean and Poincaré-ball prototype classifiers across 150 seed-replicated runs and shipped a reproducible end-to-end artifact. For the past several months I have been running activation-steering experiments on Mixtral 8x7B-Instruct on the MIT ORCD cluster, and a working draft of the conference paper that came out of those experiments is checked into the project repository.² A focused Mixture-of-Experts (MoE) follow-up to that work — Attention-Guided Router Steering (AGRS) — is planned as a *summer 2026* workshop or short-paper submission, and is what the rest of this proposal calls “the summer paper.” The MoE work is preparation, not thesis content.

Why the pivot from MoE-specific steering to scalable oversight. The summer paper establishes that attention-weighted readout can produce a more concentrated risk-difference signal on a frozen MoE language model than the upstream STEERMOE recipe does. That result is, on its own, a methods note. The thesis takes the underlying engine — contrastive readout-and-intervene on a frozen language model — and asks the harder question: *if the readout can be trusted because it was extracted from a smaller, more interpretable model, can we use it to steer and verify a larger model whose internals we cannot inspect end-to-end?* That question is scalable oversight, and it does not require an MoE architecture; dense transformers are sufficient for the proof-of-concept the thesis builds.

Why this matters. Scalable oversight is one of the load-bearing problems for the safety of advanced AI: as models become more capable than the humans who deploy them, the question of how a less-capable supervisor can verify model behavior becomes more urgent [21, 12, 11]. The dominant published approaches operate at evaluation time on the model’s *outputs* (sandwiching [7], debate [17], AI-assisted critique [31], RLAIF / Constitutional AI [3]). The thesis explores a complementary inference-time avenue: instead of asking “does the strong model’s output look acceptable?” it asks “can the strong model’s *behavior* be steered by, and verified against, a concept direction the weaker model commits to?” If yes, oversight becomes a closed loop on the model’s internal trajectory, not just its outputs.

Why this is feasible now. Three empirical developments converged in the last ~18 months. First, weak-to-strong generalization [10] demonstrated that a weaker supervisor’s training signal can elicit strong-model behavior beyond the weak supervisor’s own ability. Second, the activation-steering literature matured to the point that single-model concept directions (ITI [23], contrastive activation addition [27], representation engineering [39], sparse-autoencoder features [35]) reliably move generations on production-relevant checkpoints. Third, the cross-model representation-alignment literature [19, 4, 16] established that representations across models of similar training regime are linearly alignable. Cross-model concept transfer sits at the intersection: weak-to-strong generalization gives the *epistemic* story; activation steering gives the *mechanism*; representation alignment gives the *technical feasibility*.

¹github.com/pgrindehollevik-harvard/hyperbolic.

²Working draft and code: github.com/pgrindehollevik-harvard/MoE_attention_guided_steering (paper-submission branch).

Connection to my advisor’s research. Adit’s recent work introduces the Average Gradient Outer Product (AGOP) $M = \mathbb{E}_x[\nabla_x f(x)\nabla_x f(x)^\top]$ as a mathematical handle on *which input directions a neural network treats as informative for a given prediction* [29, 5, 30]. The cross-model concept-transfer question has a natural AGOP formulation: a transfer succeeds when the AGOP subspace of the concept on W overlaps with the AGOP subspace of the same concept on S . The mechanistic chapter measures this subspace alignment (via principal angles or the Bures–Wasserstein distance between AGOP matrices) and asks whether it predicts transfer success. This is a sharper, more falsifiable use of AGOP than the original MoE-only framing.

Future plans. After this master’s I intend to pursue research roles in AI interpretability, alignment evaluation, and post-training intervention methods, with industry research labs working on scalable oversight and behavioral steering as the primary target. The thesis is designed to produce a publication artifact and the end-to-end empirical-research experience that those programs evaluate against.

5. Goals and Expected Outcomes

1. **Protocol.** Define and implement three cross-model concept-transfer protocols — prompt curation (PC), activation correlation (AC), and probe-as-supervisor (PaS) — as drop-in alternatives to within-model steering. Each protocol takes a concept-bearing contrastive prompt set and a pair of models (W, S), and outputs a steering direction in S ’s representation space. *Success criterion:* At least two of the three protocols clear the gate in Goal 2.
2. **Empirical evaluation.** A pre-registered comparison of cross-model transfer against within- S steering as the upper-bound reference, across three model families (Pythia 1.4B/6.9B/12B; Llama-3.2 1B/3B/8B; Qwen2.5 0.5B/7B), four concept families (refusal, sycophancy, truthfulness, sentiment), and four transfer protocols (PC, AC, PaS, plus a learned linear map as a stretch goal). Sample size $n = 250$ cases per condition, five seeds. Primary metric: *Transfer Fidelity Ratio* $\text{TFR} = \text{CTR}_{\text{cross}}/\text{CTR}_{\text{within-}S}$, with concept-transfer rate (CTR) scored by an LLM judge cross-validated against a human-rater subset of 200 cases. *Success criterion:* On the refusal family, $\text{TFR} \geq 0.7$ in $\geq 2/3$ model families using at least one protocol, with non-overlapping paired bootstrap 95% CIs against random-direction control.
3. **Mechanistic analysis.** Estimate AGOP matrices M_c^W and M_c^S for each concept on W and S over the same contrastive prompt pool, and measure their principal-angle alignment. *Success criterion:* the per-(model-family, concept) correlation between AGOP subspace alignment and TFR reaches $|r| \geq 0.5$, providing a structural account of when cross-model transfer is feasible and when it is not.

Deliverables. A submitted NeurIPS or ICLR main-track paper, a thesis monograph, and an anonymized public artifact tree under CC-BY (executed notebooks, generation metadata, judge transcripts, Apptainer image, OSF pre-registration).

6. Scientific Justification

Scalable oversight is one of the foundational unsolved problems in alignment, and the published methodology is dominated by evaluation-time output-supervision. Inference-time *behavior*-supervision via cross-model concept transfer is an undertested avenue with a clean operational

claim: if a smaller, more inspectable model can provide a concept direction that meaningfully steers a larger, less inspectable one, then the oversight loop is structurally different from sandwiching or debate, and gives us a way to verify that the larger model’s response to a concept is constrained by something we have already validated on a model we trust.

The contribution is innovative in three specific ways. (i) *Direction*. Cross-model concept transfer in the activation-steering literature is essentially unexplored; the only adjacent work [10, 14] operates at training time, not inference time. (ii) *Falsifiability*. The pre-registered TFR gate against random-direction controls separates “the transfer carried real concept information” from “the transfer carried any direction.” (iii) *Mechanistic grounding*. The AGOP subspace-alignment analysis turns the empirical headline into a structural prediction: transfer should work where the concept-relevant subspaces align, and fail where they do not.

7. Literature Review

The thesis sits at the intersection of five literatures. Each is summarised briefly below; the synthesis paragraph at the end positions the contribution.

(i) Scalable oversight and amplification. The framing originates in Christiano et al. [11] (supervising strong learners by amplifying weak experts) and Leike et al. [21] (reward modeling at scale); the empirical literature has since produced sandwiching [7], AI-assisted critique [31], debate [17], recursive reward modeling, and constitutional AI / RLAIIF [3]. Cotra [12] maps the threat model the agenda addresses. All of these methods are output-supervision protocols; the thesis is complementary behavior-supervision.

(ii) Weak-to-strong generalization. Burns et al. [10] showed that a weak supervisor’s labels can train a strong student to perform meaningfully above the supervisor’s own ceiling under certain conditions, opening the question of when and why this is possible. Follow-up work has explored the failure modes [37], easy-data amplification [14], and weak-to-strong reasoning [38]. These are training-time results; the thesis asks the inference-time analogue.

(iii) Activation steering and representation engineering. The line begins with latent steering vectors [33] and matures into Inference-Time Intervention [23], Activation Addition [36], Representation Engineering [39], Contrastive Activation Addition [27], and the generalization-and-reliability audits of Tan et al. [34]. A parallel line uses sparse autoencoders to extract monosemantic feature directions [8, 35, 26] and uses linear probes to read concept-relevant directions [1, 25]. All of this work is within-model; the thesis treats these methods as *concept-extraction primitives* that the cross-model protocol calls into.

(iv) Cross-model representation alignment. Cross-model alignment of internal representations has been established empirically by Kornblith et al. [19] (centered kernel alignment), Bansal et al. [4] (model stitching), and Lenc and Vedaldi [22]. Huh et al. [16] argues for a “platonic representation” converged across scale and architecture; Park et al. [28] formalises the linear representation hypothesis at the population level. The thesis treats these results as the *enabling assumption* of cross-model concept transfer: if representations align, then a concept direction extracted from W can be expected to have a recoverable image in S .

(v) **Concept extraction, probing, and safety steering.** Probing classifiers [1, 6] and TCAV [18] established the supervised-direction toolkit. Burns et al. [9] (Contrast-Consistent Search, CCS) and Mallen et al. [24] (tuned lens) extended the toolkit to unsupervised and cross-layer settings. On the safety side, Arditì et al. [2] showed that refusal in Llama-2 is mediated by a single direction, Lee et al. [20] gave a mechanistic account of alignment-algorithm side effects, Sharma et al. [32] characterized sycophancy as a learned behavior, and Hubinger et al. [15] and Greenblatt et al. [13] built model organisms for deception-adjacent failure modes. These works supply the *concept families* the thesis steers on; the safety-skewed selection (refusal, sycophancy, truthfulness) reflects which directions the alignment community most wants oversight on.

Where the thesis fits. Five literatures, no single point in the cross-product currently addressed: the thesis is the first systematic empirical study of *cross-model concept transfer for inference-time oversight*, combining concept-extraction primitives from (iii), enabling assumptions from (iv), safety-relevant target directions from (v), the scalable-oversight motivation from (i), and the weak-supervisor framing from (ii). The AGOP analysis (Section 8.4) brings my advisor’s feature-learning framework into a setting it has not been applied to.

8. Methodology

Setup. For each model family I take a small model W and a larger model S from the same family (controlled training) and run a contrastive prompt set $\{(p_i^+, p_i^-)\}$ through both. The same prompts and the same tokenizer (within family) ensure that positions are comparable across W and S . For one cross-family control I pair W from one family with S from a different family to test whether shared training is required for transfer.

Three transfer protocols. *Prompt Curation (PC).* W scores prompts by within- W concept-direction activation; the top-scoring contrastive subset is used to extract a within- S steering direction with standard methods (CAA, ITI, RepE). The transferred object is the prompt set, not an activation. Simplest baseline; the operational interpretation is “ W tells us which prompts to use.”

Activation Correlation (AC). On a held-out calibration set, W ’s concept-direction activation per prompt is correlated with S ’s residual-stream activation at each (layer, position) cell; the highest-correlation cells become the readout target for S ’s direction. No learned map; the transfer is structural.

Probe-as-Supervisor (PaS). A linear probe is trained on W ’s activations for the concept; the probe is then applied to S ’s activations to label tokens or positions where the concept is present; S ’s steering direction is extracted with W ’s probe as the per-token signal rather than the prompt label.

A fourth, more ambitious protocol — a learned linear map between W and S ’s representation spaces — is included as a stretch goal in Phase 6 of the project plan.

Verification loop. For each steered S -generation, W scores whether the generation exhibits the target concept (using the same within- W concept-direction activation that defined the transfer). The verification score is cross-validated against (a) an LLM judge (GPT-4o, with Claude as cross-check) and (b) a 200-case human-rater subset, with reported Cohen’s κ and Spearman.

Mechanistic AGOP analysis. For each concept c and each model $M \in \{W, S\}$, estimate

$$M_c^{(M)} = \mathbb{E}_{x \sim p^+ \cup p^-} \left[\nabla_x f_c^{(M)}(x) \nabla_x f_c^{(M)}(x)^\top \right]$$

where $f_c^{(M)}$ is the logit of M ’s concept-direction probe at its chosen readout layer. Compute principal-angle alignment between the leading subspaces of $M_c^{(W)}$ and $M_c^{(S)}$, and report it per (model-family, concept) cell against TFR. The hypothesis is that TFR is monotone in subspace alignment.

Models and compute. *Model families.* Pythia 1.4B / 6.9B / 12B (clean capability-scaling axis, same training); Llama-3.2 1B / 3B and Llama-3.1 8B (production-relevant family); Qwen 2.5 0.5B / 7B (non-English-balanced family). *Compute.* A single NVIDIA A100 per generation job for the 1B–3B tier, two A100s for the 7B–12B tier; budget ~ 250 GPU-hours total on MIT ORCD `mit_normal_gpu` across the fall and spring.

Reproducibility. Every experimental step is a Jupyter notebook with a `papermill` parameters cell, run identically on a laptop in smoke mode and on the cluster via a paired `sbatch` wrapper. Determinism (temperature 0, fixed seeds), revision-SHA pinning, per-run `provenance.json`, an Apptainer image, and OSF pre-registration of the TFR gate before any cross-model-vs-random-direction comparison runs.

Timeline.

Months	Milestone
Jun–Aug 2026	Summer paper: AGRS for MoE routing-bias steering (preparatory).
Sep 2026	Reproducibility hardening; concept-extraction primitives implemented.
Oct 2026	Three transfer protocols (PC, AC, PaS) implemented and unit-tested.
Nov 2026	Pilot run on Pythia (1.4B→6.9B); pre-registered TFR gate evaluated.
Dec 2026	Refusal family across all three model families.
Jan 2027	Sycophancy and truthfulness families; verification-loop cross-validation.
Feb 2027	AGOP subspace-alignment analysis; mechanistic chapter draft.
Mar 2027	Statistics, thesis monograph, conference submission.

Risks and pre-declared pivots. If the November pilot fails the pre-registered TFR gate, the thesis pivots in one of two registered directions: (a) restrict to *intra-family* transfer only and characterize the family-gap threshold at which transfer breaks down (this turns the negative result into a structural finding), or (b) replace transfer with *joint extraction* — a single concept direction extracted to satisfy both W and S simultaneously — and pre-register the joint-extraction gate. Either pivot is described in the OSF pre-registration so it cannot be backfit.

Companion materials ([paper-submission branch](#)): the summer-paper working draft at `paper/docs/paper/main.po`, the phased revision plan with code-to-build manifest at `paper/docs/REVISION_PLAN.md`, and the notebook-as-pipeline scaffolding at `notebooks/README.md`.

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